

# Prashant Piprotar

Software Engineer | AI / ML Systems

prashplus@gmail.com | +91 95746 30231 | Bengaluru | www.prashplus.com | linkedin.com/in/prashantpiprotar | github.com/prashplus

## PROFESSIONAL SUMMARY

Software Engineer with 5+ years of production experience and an M.Tech in Computer Science from IIT Kanpur, building GPU-accelerated machine learning systems: real-time 3D reconstruction, computer vision and object detection with PyTorch and CUDA, and LLM tooling with LangChain and Model Context Protocol (MCP). Combines distributed systems and HPC expertise with full-stack delivery (FastAPI, Next.js, Docker) to ship ML products end to end.

## SKILLS

**Languages:** JavaScript, TypeScript, Python, Java, C++, SQL
**Frameworks & Libraries:** React.js, Redux, Next.js, Node.js, Express.js, Spring Boot, FastAPI, Material-UI
**Backend & APIs:** REST APIs, Microservices, WebSockets, API Design, Model Context Protocol (MCP), LangChain, LangGraph
**Databases & Caching:** MongoDB, MySQL, Redis
**Cloud & DevOps:** AWS (EC2, S3), GCP (GKE), Azure (VMs, Blob Storage), Docker, Kubernetes, CI/CD (CircleCI), Git
**Distributed Systems & HPC:** Apache Spark, OpenMPI, Ceph
**AI / ML:** PyTorch, CUDA, OpenCV, YOLO, scikit-learn, Computer Vision

## PROJECTS

**AetherScan** 2026
*Next.js, FastAPI, PyTorch, Three.js, Docker*

- Built a GPU-accelerated 3D reconstruction pipeline with Fast3R and CUDA that converts multi-view images into dense point clouds.
- Streamed point data over WebSockets as raw Float32Array buffers, bypassing React state to render 1M+ points at 60 FPS.
- Containerized the multi-service architecture with Docker Compose and NVIDIA Container Toolkit for reproducible GPU deployments.

**Real-Time Object Detection** 2025
*Python, PyTorch, CUDA, OpenCV, Streamlit*

- Developed a real-time object detection service using YOLO with CUDA acceleration for low-latency video inference.
- Built a Streamlit monitoring dashboard with live frame-rate tracking and configurable confidence thresholds for camera streams.

**Evaluation of Spark and MPI for Scientific Data Analysis** 08/2020 – 05/2021
*IIT Kanpur (M.Tech Thesis) | Apache Spark, C++, OpenMPI*

- Benchmarked OpenMPI against Apache Spark on distributed clusters, comparing throughput on CPU-, memory-, and I/O-intensive workloads.
- Analyzed performance on large scientific datasets from astrophysics, meteorology, and combustion simulations.

**Malware Detection using Machine Learning** 01/2020 – 05/2020
*IIT Kanpur | Python, scikit-learn*

- Trained and evaluated Random Forest classifiers to detect malicious binaries using static and dynamic analysis features.

## WORK EXPERIENCE

**Senior Software Engineer** 06/2023 – Present
*Enphase Energy* *Bengaluru, India*

- Designed and launched the Solargraf Public REST API and developer documentation in 4 months, reducing partner onboarding time and accelerating third-party integrations.
- Defined engineering standards across the Solargraf platform by improving API performance, expanding automated test coverage, and mentoring engineers across teams.

**Software Engineer** 08/2021 – 05/2023
*Enphase Energy* *Bengaluru, India*

- Built an automated Bill of Materials (BOM) calculator with real-time pricing integration in 3 months, increasing customer satisfaction scores by 5%.
- Optimized front-end rendering and back-end database queries across core installer tools, reducing latency and shortening quote turnaround time.

**Research Intern** 01/2018 – 04/2018
*Centre for Development of Advanced Computing (C-DAC)* *Pune, India*

- Built an open-source network emulator and integrated IDE for software-defined networking (SDN), reducing setup time for researchers developing SDN applications.
- Deployed a Ceph distributed storage cluster for the SDN Online Lab, improving fault tolerance, high availability, and data retrieval speed.

## EDUCATION

**M. Tech. in Computer Science and Engineering** 07/2019 – 06/2021
*Indian Institute of Technology, Kanpur* *GPA: 8.6/10*

**B. Tech. in Computer Engineering** 07/2014 – 05/2018
*CSPIT, Charotar University of Science and Technology* *GPA: 9.04/10*

**XII (C.B.S.E.)**
*K.V. No. 2, Jamnagar* *81%*

**X (C.B.S.E.)**
*K.V. No. 2, Jamnagar* *GPA: 10/10*

## CERTIFICATIONS

**Introduction to LangGraph** — LangChain Academy
**Essential Cloud Infrastructure: Foundation** — Google Cloud
**Essential Cloud Infrastructure: Core Services** — Google Cloud
**MTA: Windows Server Administration Fundamentals** — Microsoft 2017
**Cisco Certified Network Associate Routing and Switching (CCNA)** — Cisco 07/2015 – 04/2017

## AWARDS & SCHOLARSHIPS

**All India Rank 422, GATE** 01/2019

## LANGUAGES

**English** (Full Professional Proficiency), **Hindi** (Native), **Gujarati** (Native)